Sandeep Kundalwal

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Experience

PropertyGuru Group, Singapore Sept 2024 – Present
Software Engineer — Full-Stack *Bangalore, Karnataka*

- Cut property research time by ∼**91.0%** (*Best Business Impact Award, Aug 2025*) by architecting an enrichment pipeline for **18K+ properties**: 12 parallel agent calls per property, cross-validated across two LLMs (*Gemini*, *GPT*) with confidence-scored field-level reconciliation.
- Drove a **40%** repeat-engagement uplift and scaled to **300K+ notifications/month** at **98.43%** delivery by solely building a multi-channel (email/WhatsApp) event-driven SQS notification system that replaced a manual batch workflow for **44K users**.
- Restored leadership metrics feeding a live **SGD 300K campaign** in **4 days** by tracing inflated numbers to a legacy `WRITE_APPEND` pipeline and building an idempotent Airflow ELT fix (staging + `MERGE`) with zero legacy code changes; backfilled 12 tables, removing ∼**1M duplicate rows**.
- Built the **BFF layer** for My Property from scratch, introducing a **localization foundation** (static text served from BFF instead of hardcoded app strings) now **adopted platform-wide** across PropertyGuru — by aggregating and transforming calls across multiple backend services via **GraphQL-based selective fetching**.
- Built the My Property app's **agent-proposals system** — dashboard preview, filterable/paginated full-list screen, and bulk-enquiry screen — synced via **Jotai, MMKV, and WatermelonDB** to avoid backend calls on every property switch across up to **4 properties/user**.
- Redesigned backend data models and APIs to support multi-property accounts, then extended the system with an idempotent, transactional *RabbitMQ* pipeline for cross-channel (email/WhatsApp) identity merges, removing the 4-property limitation for ∼**9,800 of 130K+ users** via user-guided soft deletion; delivered the end-to-end solution solo in **under 2 weeks** ahead of a hard launch deadline.
- Developed a geo-spatial analytics service on *H3 hexagonal indexing* (Res 8, ∼600m) that shows nearby listing view trends to homeowners; cached hex-level view aggregates in Redis for a **97.87%** hit rate, **63%** lower p99 latency, and **500+** peak RPS.

IT University of Copenhagen, Denmark Jun 2024 – Aug 2024
Software Engineer Intern *Remote*

- Built a *Java/Spring Security*-based authentication and authorization service (PEP, PDP, ABAC) for a satellite-driven irrigation advisory app serving Himalayan farmers — enforced attribute-based access control on all API endpoints and containerized the full auth stack using *Docker*, securing communication between mobile frontend and backend; part of a research collaboration between ITU Copenhagen, IIT Mandi, and DHI GRAS (Denmark).

Technical Skills

Languages: Java, TypeScript, JavaScript, Python
Frontend: React.js, Next.js, React Native
Backend: NodeJS, NestJS, GraphQL, REST APIs
Databases: MongoDB, MySQL (Relational), BigQuery, Redis (AWS ElastiCache)
Messaging & Streaming: Apache Kafka (MSK), RabbitMQ, AWS SQS (Event-Driven Architecture)
Cloud & Infra: AWS (S3, EC2), Terraform, Docker, Airflow (ETL), Datadog, Kibana
Engineering: Automated Testing (**80%+** Coverage), SDLC, Event-Driven Architecture, AI-Assisted Development (Claude Code, GitHub Copilot)

Education

Indian Institute of Technology Mandi
Master of Technology in Computer Science *CGPA: 8.17*

Rajasthan Technical University, Kota
Bachelor of Technology in Computer Science *CGPA: 8.72*

Achievements

- **Best Business Impact Award** — *PropertyGuru Group Singapore, Aug. 2025 (Company Wide).*
- **IEEE Conference Publication (ICCCNT)** — *Co-authored a research paper using eye-tracking to analyze user behavior on AI vs. human-crafted phishing emails, identifying critical gaps between visual attention patterns and detection accuracy.*
- **Gate CS '22** — *99.08 percentile, top 1% among 100K+ candidates nationwide.*
- Solved **940+** DSA problems on Leetcode.